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RESEARCH INTERESTS

- Plant Hormone Crosstalk & Signal Transduction, Functional Genomics & Stress Physiology

EDUCATION

- **Ph.D. in Horticulture — Virginia Tech, USA (2026)**
 - *Dissertation:* Physiological and Molecular Approaches Revealing Mechanisms of Fruit Abscission and Pre-Harvest Drop Management in Apple (*Malus × domestica* Borkh.)
 - *Advisor:* Dr. Sherif M. Sherif
 - *Location:* Alson H. Smith Jr. Agricultural Research and Extension Center (AREC), Winchester, VA
- **M.S. in Plant Pathology — Bangladesh Agricultural University, Bangladesh (2017)**
 - *Thesis:* Molecular Detection of *Candidatus Liberibacter asiaticus* Causing Citrus Greening in Bangladesh
 - *Advisor:* Dr. Md. Rashidul Islam
- **B.Sc. in Agriculture — Bangladesh Agricultural University, Bangladesh (2009)**

RESEARCH & PROFESSIONAL APPOINTMENTS

- **Post-doctoral Associate — Department of Plant Science and Landscape Architecture, University of Maryland, College Park, MD** Sep 2026–Present
 - Lead analytical chemistry, biochemical assays, metabolomics, and fruit quality profiling (GC-MS, LC-MS, HPLC) evaluating tree fruit physiology and post-harvest storability
 - Conduct RNA-seq transcriptomic expression profiling, qRT-PCR assays to decipher post-harvest fruit quality targeting hormonal pathways and stress-responsive candidate genes under varying temperature regimes.
 - Coordinate research with agricultural experiment stations and industry stakeholders; mentor graduate and undergraduate student researchers in laboratory and post-harvest analytical procedures.
- **Adjunct Part-time Faculty / Summer Researcher — Alson H. Smith Jr. AREC, Virginia Tech, Winchester, VA** Jun 2026–Aug 2026
 - Investigated physiological mechanisms and hormonal cascades regulating abscission zone activation and pre-harvest drop in tree fruit systems.
 - Contributed preliminary data, physiological rationale, and literature syntheses to support competitive federal and state grant applications.
- **Graduate Research Assistant — Alson H. Smith Jr. AREC, Virginia Tech, Winchester, VA** Aug 2022–May 2026
 - Investigated the molecular and hormonal crosstalk (ethylene vs. auxin) regulating abscission signaling networks, successfully decoupling fruit drop from ripening through coordinated plant growth regulator (PGR) regimes.

- Performed nucleic acid extractions (CTAB/spin columns), cDNA synthesis, and extensive qRT-PCR assays for target transcript profiling; conducted DNA methylation analysis.
 - Monitored canopy phenology, fruit maturation indices; quantified internal ethylene dynamics via gas chromatography and evaluated reactive oxygen species (ROS), anthocyanin accumulation, and chlorophyll degradation via spectrophotometry and colorimetric systems.
 - Collaborated with multidisciplinary research teams to develop orchard management strategies mitigating climate extremes and spring frost damage in perennial fruit crops through hormonomics to delay bloom (peaches and apples).
 - Analyzed high-dimensional transcriptomic, biochemical, and physiological datasets using R and prepared publication-standard scientific visualizations.
- **Research Specialist — Horticulture & Plant Pathology, Bangladesh** Mar 2013–Jul 2022
Agricultural Research Institute (BARI), Gazipur, Bangladesh
 - Designed and executed multi-environment field trials and laboratory experiments evaluating genetic diversity, germplasm performance, and biotic/abiotic stress management across major food and horticultural crops.
 - Established standard molecular diagnostic platforms (conventional PCR, qPCR, gel electrophoresis) for high-consequence plant bacterial and viral pathogens (*Ralstonia solanacearum*, *Candidatus Liberibacter asiaticus*, Citrus tristeza virus) under ISO/IEC 17025 laboratory quality frameworks.
 - Screened germplasm and mutant lines for disease resistance and agronomic yield stability in multi-location field trials.
 - Analyzed yield, quality, and disease incidence data; authored peer-reviewed journal articles and comprehensive technical reports for extension agents, growers, and agricultural ministries.
 - **Research Assistant — Citrus Development Project, BARI, Bangladesh** Aug 2015–Feb 2017
 - Executed national field surveys evaluating Huanglongbing (HLB / citrus greening) incidence, pathogen spread, and orchard decline across commercial citrus orchards.
 - Developed molecular diagnostic workflows for citrus rootstocks and scions, publishing findings in peer-reviewed journals and technical guides for researchers, commercial nursery managers and growers.

TECHNICAL SKILLS & EXPERTISE

- **Horticultural & Tree Fruit Physiology:** Experimental orchard design, canopy management, tree fruit phenology tracking, plant growth regulator (PGR) screening, bloom delay/frost mitigation, and harvest maturity determination.
- **Molecular Biology:** Total RNA/DNA isolation from plant tissues (CTAB/Kit), reverse transcription, quantitative real-time PCR (qRT-PCR), primer design, DNA methylation analysis.
- **Plant Tissue Culture & In Vitro Systems:** Aseptic micropropagation, callus induction, organogenesis, media preparation, and maintenance of sterile culture lines.
- **Field & Analytical Instrumentation:** Gas chromatography (internal ethylene concentration), spectrophotometric assays (ROS, enzyme kinetics), titrator (acidity, pH), colorimetry, chlorophyll fluorescence, and fruit firmness texture analyzer and penetrometers.

- **Pathology & Diagnostics:** Nationwide disease survey methodology, pathogen epidemiology, molecular diagnostic screening for *Candidatus Liberibacter asiaticus* (HLB).
- **Data Science & Biometrics:** RNA-seq differential expression analysis (DESeq2, UpSetR), transcript abundance quantification, basic sequence alignment and phylogenetics (BLAST, CLUSTAL-W, MEGA, Geneious), functional enrichment (DAVID) and reproducible statistical modeling in R and JMP (ANOVA, linear mixed models, multivariate clustering, PCA).

PEER-REVIEWED PUBLICATIONS

Journal Publications

- **Tipu MMH**, Jahed KR, Sherif SM (2025) Integrated application of ACC and AVG mitigates pre-harvest fruit drop and enhances red coloration in apple. *Plant Growth Regulation*, 105:1289-1305. DOI: [10.1007/s10725-025-01344-0](https://doi.org/10.1007/s10725-025-01344-0)
- Jahed KR, Saini AK, **Tipu MMH**, Das PR, Erland L, Sherif SM (2025) Hormonomics-driven bloom regulation and delay: Enhancing climate resilience in peach. *Plant Stress*, 18:101101. DOI: [10.1016/j.stress.2025.101101](https://doi.org/10.1016/j.stress.2025.101101)
- **Tipu MMH**, Sherif SM (2024) Ethylene and its crosstalk with hormonal pathways in fruit ripening: mechanisms, modulation, and commercial exploitation. *Frontiers in Plant Science*, 15:1475496. DOI: [10.3389/fpls.2024.1475496](https://doi.org/10.3389/fpls.2024.1475496)
- Rezvi HUA, Tahjib- Ul- Arif M, Azim MA, Tumpa TA, **Tipu MMH**, Najnine F, Dawood MFA, Skalicky M, Brestič M (2023) Rice and food security: Climate change implications and the future prospects for nutritional security. *Food and Energy Security*, 12: e430. DOI: [10.1002/fes3.430](https://doi.org/10.1002/fes3.430)
- Ema IJ, Monika MA, Khan AU, **Tipu MMH**, Faruk MR, Tarapder SA, Adnan M (2022) A Review on the Management of Country Bean (*Lablab purpureus* L.) Diseases in Bangladesh. *The Journal of Agricultural Sciences - Sri Lanka*, 17(3): 388-415p. DOI: [10.4038/jas.v17i3.9920](https://doi.org/10.4038/jas.v17i3.9920)
- Akhter MS, **Tipu MMH**, Rahman MS, Islam R, Faruk MI, Rahman MM, Nakahara KS (2022) First report of *Citrus tristeza* virus in Bangladesh. *Australasian Plant Disease Notes*, 17:12. DOI: [10.1007/s13314-022-00457-z](https://doi.org/10.1007/s13314-022-00457-z)
- Islam MM, Naznin S, Naznin A, Uddin MN, Amin MN, Rahman MM, **Tipu MMH**, Alsuhaibani AM, Gaber A, Ahmed S (2022) Dry Matter, Starch Content, Reducing Sugar, Color and Crispiness Are Key Parameters of Potatoes Required for Chip Processing. *Horticulturae*, 8(5): 362. DOI: [10.3390/horticulturae8050362](https://doi.org/10.3390/horticulturae8050362)
- **Tipu MMH**, Jahan R, Rahman J, Rahman MM, Islam MA, Apu, MMRB (2021) Development of efficient integrated management package against sweet potato weevil (*Cylas formicarius* [Fabricius, 1798]). *Acta Agriculturae Slovenica*, 117(4): 1-4p. DOI: [10.14720/aas.2021.117.4.2066](https://doi.org/10.14720/aas.2021.117.4.2066)
- **Tipu MMH**, Masud MM, Jahan R, Baroi A, Hoque AKMA (2021) Identification of citrus greening based on visual symptoms: A grower's diagnostic toolkit. *Heliyon*, 7(11): e08387. DOI: [10.1016/j.heliyon.2021.e08387](https://doi.org/10.1016/j.heliyon.2021.e08387).
- Hoque AKMA, Chowdhury HT, **Tipu MMH**, Biswas S, Islam MR, Rahman MM (2021) Fruit traits diversity in Burmese grape (*Baccaurea ramiflora*) through multivariate analysis. *Agronomy Journal*, 113: 4753-4761p. DOI: [10.1002/agi2.20904](https://doi.org/10.1002/agi2.20904)
- Amin MN, Rahman MM, Naznin S, Alam MK, **Tipu MMH**, Prodhan MZH, Islam MM, Kundu BC (2021) Tuber Yield and Stability Assessment of Potato Genotypes in Bangladesh. *Indian Journal of Agricultural Research*. DOI: [10.18805/IJARE.A-607](https://doi.org/10.18805/IJARE.A-607)
- Elahi F, Islam M, Alam M, Islam N, **Tipu MMH**, Khatun F, Chowdhury SR, Sarker S (2021) First report of *Athelia rolfsii* on jackfruit (*Artocarpus heterophyllus*). *Australasian Plant Disease Notes*, 16:17. DOI: [10.1007/s13314-021-00430-2](https://doi.org/10.1007/s13314-021-00430-2)

- **Tipu MMH**, Jahan R, Rahman J, Riad MI, Rahman MM, Nabi KME (2021) Status of major diseases of brinjal and tomato in charland of Jamalpur and Sherpur districts of Bangladesh. *Plant Science Today*, 8(1): 161-165p. DOI: [10.14719/pst.2021.8.1.988](https://doi.org/10.14719/pst.2021.8.1.988)
- Islam MR, Islam MN, Alam MZ, Hossain MM, Hussien MAM, **Tipu MMH** (2020) Chemical inducers, nutrient management, guava intercropping and insecticides can reduce Huanglongbing incidence and severity in Sweet orange. *Archives of Agriculture and Environmental Science*, 5(4): 436-446p. DOI: [10.26832/24566632.2020.050401](https://doi.org/10.26832/24566632.2020.050401)
- **Tipu MMH**, Rahman MM, Islam MM, Elahi F-E, Jahan R, Islam MR (2020) Citrus Greening Disease (HLB) on Citrus reticulata (Mandarin) Caused by *Candidatus Liberibacter asiaticus* in Bangladesh. *Physiological and Molecular Plant Pathology*, 112:101558. DOI: [10.1016/j.pmpp.2020.101558](https://doi.org/10.1016/j.pmpp.2020.101558)
- Kundu BC, Amin MN, Mostofa M, Naznin S, Ali MM, Rashid MH, **Tipu MMH**, Anwar MB, Ahmed NU, Rahman MM, Alam MK, Prodhan MZH, Rashid MH (2020) Selection of Exotic Potato Genotypes for Export and Processing Purposes in Bangladesh. *Azarian Journal of Agriculture*, 7(4): 112-120p. DOI: [10.29252/azarinj.036](https://doi.org/10.29252/azarinj.036)
- Yasmin M, Rahman MA, Shikha FS, Rahman MS, Rahman J, **Tipu MMH** (2020) Effect of Mulch on Soil Temperature, Soil Moisture Conservation and Yield of Chilli. *Journal Clean WAS*, 4(1): 36-39p. DOI: [10.26480/jcleanwas.01.2020.36.39](https://doi.org/10.26480/jcleanwas.01.2020.36.39)
- **Tipu MMH**, Islam MR, Azmatullah M (2017) *Candidatus Liberibacter asiaticus* Causing Citrus Huanglongbing on Citrus sinensis in Bangladesh. *Journal of Plant Pathology*, 99(1): 293p. [\[Link\]](#)
- Rahman MM, Haque ME, Parveen K, **Tipu MMH**, Islam MS (2016) Effect of Different Sources of Nutrient and Hormones on Growth and Yield of Tomato. *Journal of Bioscience and Agriculture Research*, 11(01): 936-941p. DOI: [10.18801/jbar.110116.114](https://doi.org/10.18801/jbar.110116.114)
- **Tipu MMH**, Amin M, Dhar M, Alam MA (2014) Effects of Mulching on Yield and Quality of Tomato Varieties. *Research & Reviews: Journal of Agricultural Science and Technology*, 3(3): 12-14p. [\[Link\]](#)

Book Chapters

- **Tipu MMH**, Baroi A, Rana J, Islam S, Jahan R, Miah MS, Asaduzzaman M (2021) Potential Applications of Nanotechnology in Agriculture: A Smart Tool for Sustainable Agriculture. In: Asaduzzaman M, Afroz M (Eds.), *Agricultural Development in Asia – Potential Use of Nano-Materials and Nano-Technology*. IntechOpen, London, UK. DOI: [10.5772/intechopen.101142](https://doi.org/10.5772/intechopen.101142)

EXTENSION PUBLICATIONS & GROWER DELIVERABLES

Peer-reviewed extension publications

- **Tipu MMH**, Jahed KR, Sherif SM (2025) Naphthaleneacetic Acid: A Key Tool for Reducing Preharvest Fruit Drop and Sustaining Apple Red Coloration. Virginia Cooperative Extension, Virginia Tech. SPES-717NP.

Extension Technical Reports & Grower Bulletins

- **Tipu MMH**, Jahed KR, Sherif SM (2025) Evaluation of ACC and AVG Combination as a New Tool for Managing Pre-Harvest Fruit Drop and Improving Red Color in 'Honeycrisp' Apples. Virginia Cooperative Extension, Virginia Tech. SPES-712NP.

EXTENSION AND ACADEMIC PRESENTATIONS & GROWER WORKSHOPS

Extension & Stakeholder Demonstrations

- **University of Maryland Extension (UME) (2026-present):** Engaged tree fruit growers on harvest maturity tracking, post-harvest management techniques, and PGR trial observations.
- **Virginia Cooperative Extension (VCE) (2025):** Conducted live field and canopy management demonstrations during AREC Open House & Field Day, presenting applied PGR protocols to commercial fruit growers, crop consultants, and industry stakeholders.
- **BARI On-Farm Research and Extension (2014-19):** Coordinated and conducted regional training workshops on horticultural crop management and plant disease diagnosis for regional agricultural extension officers and commercial producers.

Academic & Conference Presentations

- Decoupling of Fruit Abscission from Fruit Ripening for Controlling Pre-Harvest Fruit Drop in Apple. 2026 Cornell Corteva Plant Breeding Symposium, Ithaca, NY, USA (2026).
- Decoupling Fruit Abscission from Ripening to Control Pre-Harvest Fruit Drop in Apple. 10th Annual University of Florida Plant Science Symposium, Gainesville, FL, USA (2026).
- Overcoming the Pre-Harvest Fruit Drop vs. Fruit Color Trade-off in 'Honeycrisp' Apples Using Combined Plant Growth Regulators. ASHS 2025 Annual Conference, New Orleans, LA, USA (2025).
- Synergistic PGR Application for Strategic Balancing of Red Color Enhancement and Fruit Drop Control in Apples. 100th Annual Cumberland-Shenandoah Fruit Workers Conference, Martinsburg, WV, USA (2024).
- Achieving Strategic Balance in Red Color Enhancement and Fruit Drop Control in Apple through Synergistic PGR Applications. 2024 Southeast Professional Fruit Workers Conference, Auburn, AL, USA (2024).
- Decoupling Fruit Abscission from Ripening to Minimize Pre-Harvest Fruit Drop in Apple. CALS 2026 Research Summit, Blacksburg, VA, USA (2026).
- Harnessing Plant Growth Regulators for Exquisite Fruit Coloration and Strategic Fruit Drop Reduction in Apple. Translational Plant Sciences Symposium 2024, Blacksburg, VA, USA (2024).
- Molecular detection and confirmation of citrus greening causal agent *Candidatus L. asiaticus* in Bangladesh. 1st Congress of the International Society for Citrus Huanglongbing, Liberibacter and other Phloem-limited Bacterial Pathosystems. Ganzhou, Jiangxi, China (2018).

GRANTS, FELLOWSHIPS & COMPETITIVE AWARDS

- P. Howard & Betsy Massey Horticulture Scholarship, Virginia Tech, USA (2024)
- Student Travel Award: 10th Annual Plant Science Symposium, University of Florida, USA (2026)
- Symposium Travel Grant: 2026 Cornell Corteva Plant Breeding Symposium, Cornell University, USA (2026)
- GPSS Travel Award: GPSS Travel Fund (Summer 2025), Virginia Tech, USA (2025)
- OKP Scholarship, Wageningen Centre for Development Innovation, Netherlands (2020)
- NEB Molecular Biology Summer Scholarship, New England Biolabs, USA (2020)
- International Congress Travel Award: 2018 ISCHLP CONGRESS I, Ganzhou, Jiangxi, China (2018)
- Citrus Development Fellowship for graduate (MS) program, BARI, Bangladesh (2015)

LEADERSHIP, SERVICE, OUTREACH & UNIVERSITY ENGAGEMENT

- **Graduate Student Grant Reviewer**, GPSS Travel Fund Committee, Virginia Tech (2025). Evaluated graduate proposals, budgets, and research abstracts to allocate competitive university travel awards.

- **Judge**, Virginia Governor's School for Agriculture Global Seminar Symposium (2025). Assessed and provided technical feedback on youth research presentations.
- **Laboratory & Research Mentor**, University of Maryland & Virginia Tech (2022–Present). Mentored undergraduate students and seasonal technicians in orchard trial layout, harvest maturity assessments, and analytical instruments.
- **Peer Reviewer**, Multiple High Impact Journals (2020–2025). Served as an invited peer reviewer for international horticultural and plant science journals, critically evaluating manuscripts for experimental rigor, methodological soundness, and scientific contribution.
- **Donor Motivation & Event Coordinator**, BADHAN Blood Donation Organization, Bangladesh Agricultural University (2005–2007). Organized blood donation events and motivated volunteers.

PROFESSIONAL MEMBERSHIPS

- American Society for Horticultural Science
- Bangladesh Phytopathological Society
- Bangladesh Society for Horticultural Science

REFERENCES

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